Mini Project Lab Report (UNIT 3)

1. Attendance Tracker

Problem Statement & Objectives

Develop a Python-based attendance tracker for a class using lists and dictionaries. The application stores student names and attendance status, enables record updates, displays attendance details in a clean format, and calculates attendance percentages to identify students below a specified threshold.

Code:

```
class AttendanceTracker:
    def __init__(self):
        # Dictionary structure: { student_name: [list of boolean attendance status] }
        self.records = {}

    def add_student(self, name):
        """Adds a new student to the system."""
        if name in self.records:
            print(f"Warning: Student '{name}' already exists.")
        else:
            self.records[name] = []
            print(f"Student '{name}' added successfully.")

    def mark_attendance(self, name, status_list):
        """Adds or updates attendance records (True for Present, False for Absent)."""
        if name not in self.records:
            print(f"Error: Student '{name}' not found.")
            return
        self.records[name].extend(status_list)

    def calculate_percentage(self, name):
        """Calculates attendance percentage for a given student."""
        if name not in self.records or not self.records[name]:
            return 0.0
        total_classes = len(self.records[name])
        attended = sum(1 for status in self.records[name] if status)
        return (attended / total_classes) * 100

    def display_records(self):
        """Displays attendance summary in an organized table format."""
        print("\n" + "=" * 65)
        print(f"{'Student Name':<20} | {'Classes Attended':<18} |{'Attendance%':<12}")
        print("=" * 65)
```

```
        for name, status_list in self.records.items():
            total = len(status_list)
            attended = sum(1 for s in status_list if s)
            pct = self.calculate_percentage(name)
            print(f"{name:<20} | {f'{attended}/{total}':<18} | {pct:>11.2f}%")
        print("=" * 65)

    def identify_low_attendance(self, threshold=75.0):
        """Identifies students falling below a specified percentage threshold."""
        print(f"\n[Alert] Students with Attendance Below {threshold}%:")
        low_att = []
        for name in self.records:
            pct = self.calculate_percentage(name)
            if pct < threshold:
                low_att.append((name, pct))
                print(f" - {name}: {pct:.2f}%")
        if not low_att:
            print(" None. All students meet the threshold requirement.")
        return low_att


# Demo / Test Scenarios
if __name__ == "__main__":
    tracker = AttendanceTracker()

    # 1. Add Students
    students = ["Alice Smith", "Bob Jones", "Charlie Brown", "Diana Prince"]
    for s in students:
        tracker.add_student(s)

    # 2. Mark Attendance across 10 lectures
    tracker.mark_attendance("Alice Smith", [True, True, True, True, True, True, True,
True, True, False]) # 90%
    tracker.mark_attendance("Bob Jones", [True, False, True, False, True, False, True,
False, False, True]) # 50%
    tracker.mark_attendance("Charlie Brown", [True, True, True, True, True, True,
True, False, True, True]) # 90%
    tracker.mark_attendance("Diana Prince", [False, True, False, True, False, True,
True, False, True, False]) # 50%

    # 3. Display Detailed Summary
    tracker.display_records()

    # 4. Check Low Attendance (< 75%)
    tracker.identify_low_attendance(threshold=75.0)
```

1.3 Execution Output Log

```
Student 'Alice Smith' added successfully.
Student 'Bob Jones' added successfully.
Student 'Charlie Brown' added successfully.
Student 'Diana Prince' added successfully.

================================================================
Student Name         | Classes Attended   | Attendance %
================================================================
Alice Smith          | 9/10               |       90.00%
Bob Jones            | 5/10               |       50.00%
Charlie Brown        | 9/10               |       90.00%
Diana Prince         | 5/10               |       50.00%
================================================================

[Alert] Students with Attendance Below 75.0%:
 - Bob Jones: 50.00%
 - Diana Prince: 50.00%
```